

Project Workbench (PWB)

High-Level Architecture (HLA) Document

Document Metadata Attribute	System Architectural Specification
Document Type	High-Level Architecture (HLA) Specification
Version & Status	Version 1.0 — Official Approved Baseline
Target System	Project Workbench (PWB) Centralized Governance Platform & Employee Hub
Security & Session Model	BFF Secure HttpOnly Cookie Boundary + Downstream Bearer Token & Role-Based Access Control

1. Introduction & Executive Context

This document describes the High-Level Architecture (HLA) of the **Project Workbench (PWB)** — a centralized Project Governance, Monitoring, and Delivery Management platform for MPOne Limited, integrating seamlessly with **MPOEmployeeHub** for enterprise identity and Single Sign-On (SSO).

1.1 Business Context & Strategic Objectives

PWB serves as the single source of truth for executive project oversight, delivering a 360-degree view across:

Functional Governance Scope Area	Functional Governance Scope Area
• Government Department Projects	• Financial & Operational KPIs
• Citizen Service Initiatives	• Milestones & Planning
• e-Governance Programs	• Alerts & Notifications
• UPP Network & District Performance	• Reporting, Analytics & Power BI
• Vendor & Partner Management	• Project Governance & Audit
• Risk & Compliance Monitoring	• Executive Dashboards & SLA Oversight

1.2 Architecture Scope & Summary

PWB follows a six-layer architecture where React provides the user interface, Next.js provides the browser-facing BFF, YARP acts as the common gateway, MPOEmployeeHub provides SSO and identity, .NET APIs provide PWB business functionality, and MSSQL provides persistence.

Layer	Layer Name	Technology	Primary Architectural Responsibility
1	React Frontend UI	React UI (Vite / SPA)	Browser-facing user interface where users interact with PWB.
2	Next.js BFF	Next.js	Backend For Frontend — manages session boundary & HttpOnly cookie.
3	YARP Gateway (.NET)	.NET / YARP	Common security & traffic gateway — routing, rate limiting, auth check.
4	MPOEmployeeHub SSO	Identity Platform	User authentication, Single Sign-On, JWT token issuance & lifecycle.
5	PWB Backend APIs	.NET / ASP.NET Core	All PWB business logic — Departments, Projects, Milestones, Risk, Finance.

6	MSSQL Server	Microsoft SQL Server	Persistent data storage — business records, transactions, audit logs
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2. Six-Layer Platform Architecture & Component Interaction

The diagram and interaction matrix below illustrate the complete 6-layer platform workflow covering the interaction between the React Frontend, Next.js BFF, YARP Gateway, MPOEmployeeHub SSO, .NET APIs, and MS SQL Server, along with key authentication, token, API, callback, and database flows.

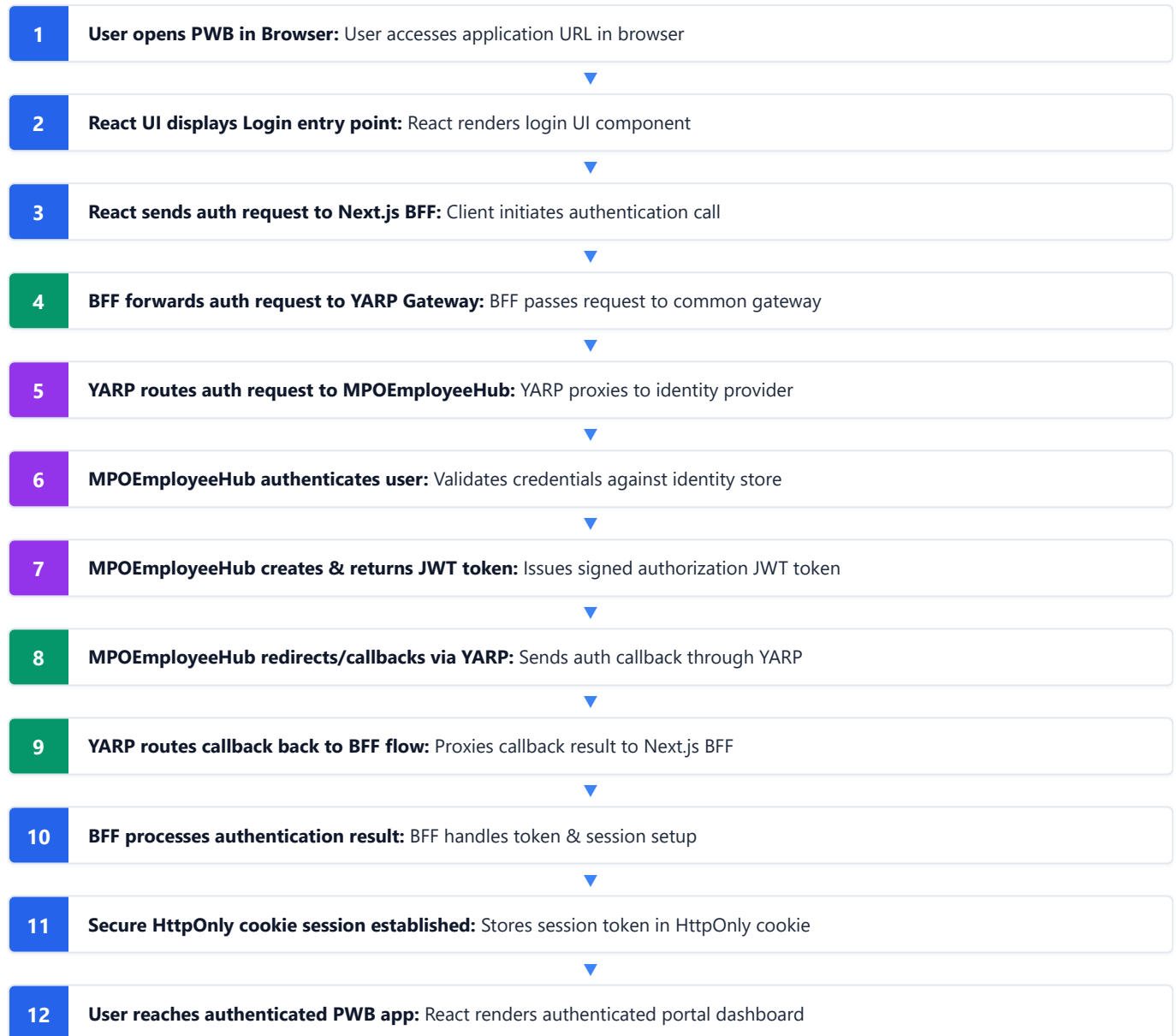


2.1 Core Security & Communication Protocol Rules

- Zero-Trust Browser Boundary:** Access tokens and refresh tokens are never delivered to or stored inside the browser's localStorage or sessionStorage.
- Strict Reverse Proxy Ingress:** Direct ingress to Layer 4 (EmployeeHub) and Layer 5 (Backend APIs) is forbidden. All external network ingress routes through Layer 3 (YARP Gateway).
- End-to-End TLS Encryption:** All transport channels across edge and internal microservices enforce HTTPS/TLS 1.3 encryption.

3. Login & Authentication Workflow

MPOEmployeeHub is the sole owner of authentication and identity. The user's login request reaches MPOEmployeeHub through the common YARP Gateway, which returns the callback through YARP to establish the BFF secure HttpOnly cookie session.



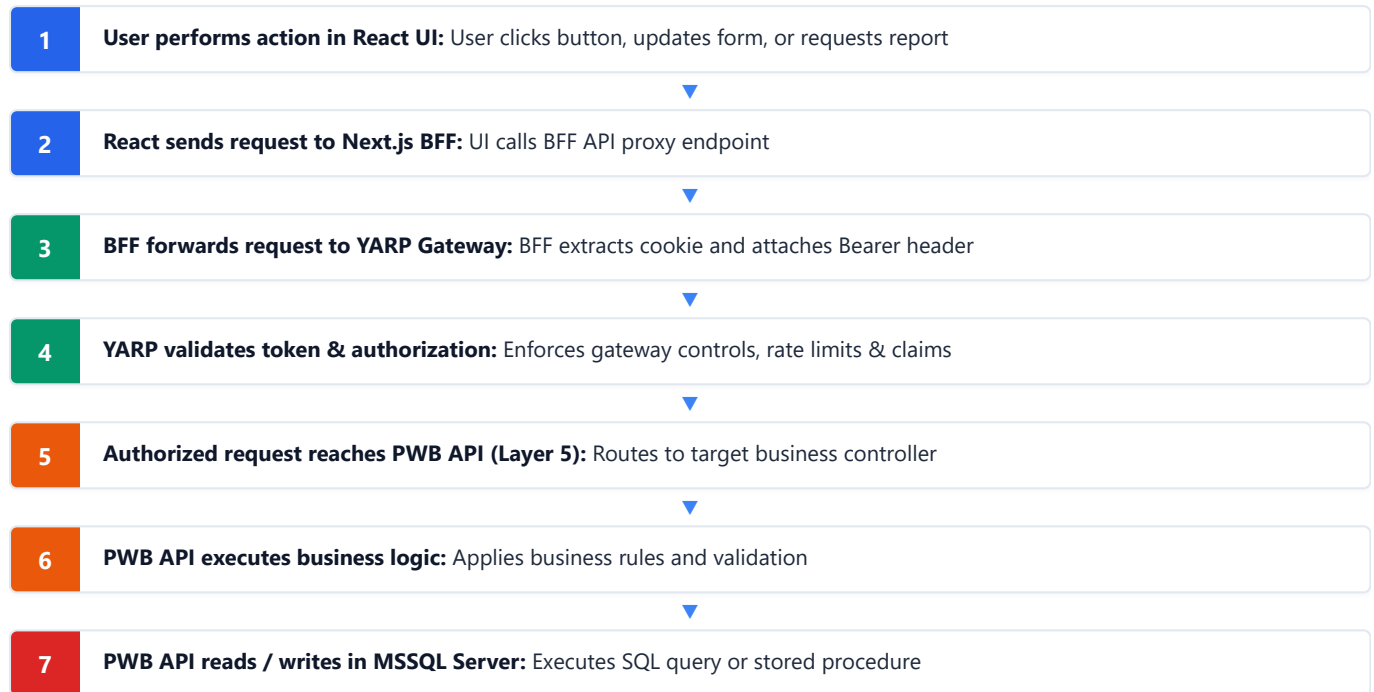
[KEY ARCHITECTURE PRINCIPLES — LOGIN]

- The browser never directly authenticates against MPOEmployeeHub without passing through the defined architecture.
- YARP is the common gateway through which all authentication traffic passes.
- The JWT / token lifecycle is owned by MPOEmployeeHub — not by the browser or React UI.
- The BFF establishes a secure HttpOnly cookie to maintain the browser session.

4. Authenticated Functional / API Workflow

After login, all user-initiated actions follow this request-response cycle. Layer 5 (PWB Backend APIs) is responsible for serving all PWB business functionality. Authorization is enforced at YARP Gateway before any business API is reached.

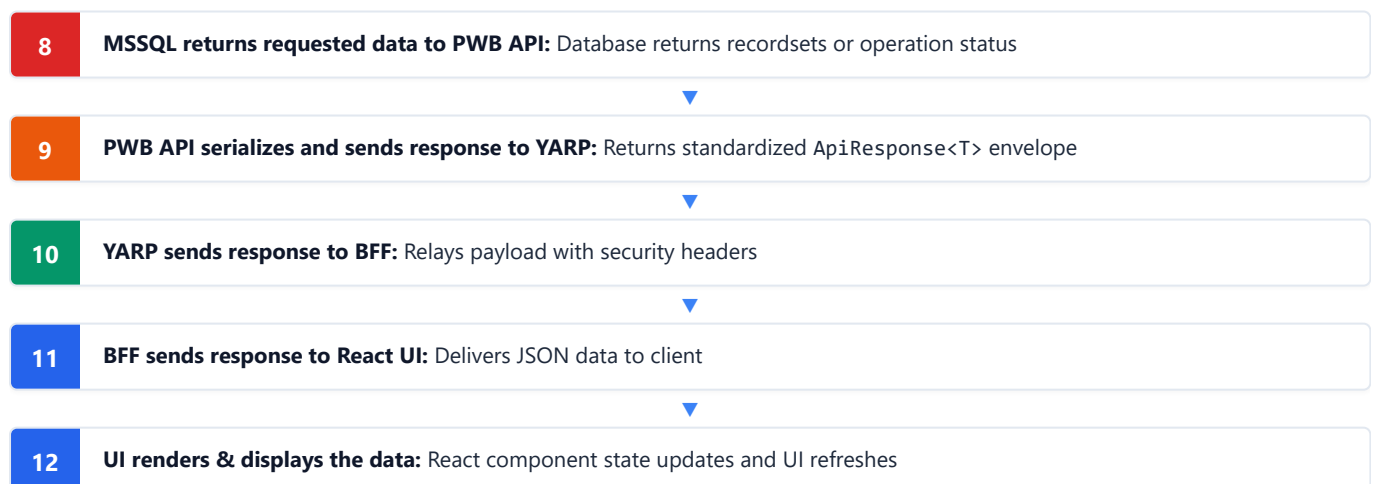
4.1 Request Execution Path (User → Database)



Layer 5 — PWB Backend API Business Domains (Illustrative Examples)

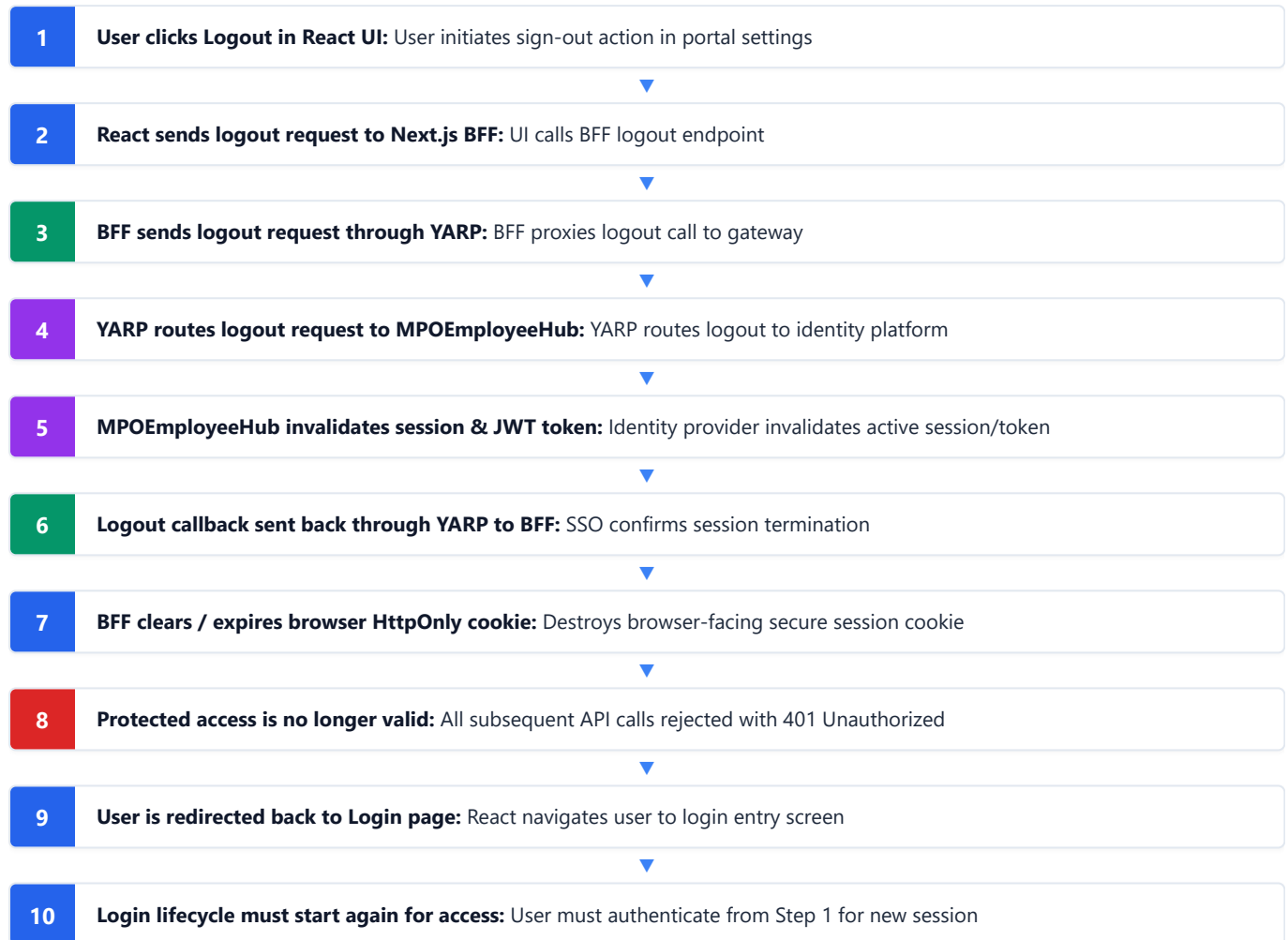
Project Portfolio | Milestones & Planning | Executive Risk | Finance & Budget | UPP Monitoring | Citizen Service | Compliance & Audit | Reporting & Analytics

4.2 Response Path (Database → User)



5. Logout & Session Invalidation Workflow

Logout is not simply hiding the UI. Authentication and session validity are invalidated at the identity layer (**MPOEmployeeHub**). Once invalidated, the user's previous session cannot be used to resume protected operations.



[SECURITY STATEMENT — LOGOUT INVALIDATION]

After logout, the user's previous authenticated session / token cannot be used to continue protected PWB operations. Protected access is invalidated at MPOEmployeeHub and BFF cookie level, forcing a complete new login lifecycle for subsequent access.

6.2 Technology Stack Summary

Component	Technology	Architectural Role & Operational Notes
Frontend UI	React UI (Vite)	Browser-based SPA UI, component-driven dashboard & interactive forms.
BFF Layer	Next.js	Server-Side BFF, session handling, HttpOnly cookie boundary.
API Gateway	.NET / YARP	Centralized reverse proxy, routing, rate limiting, security headers.
Identity Provider	MPOEmployeeHub SSO	Existing enterprise identity platform, login, JWT token lifecycle.
Business APIs	.NET / ASP.NET Core	Modular microservice APIs owning PWB business functionality.
Database	Microsoft SQL Server	Relational persistence, SSMS administration, transactions & audit logs.
Communication	HTTPS / REST	Secure encrypted communication across all inter-service layers.

6.3 Architecture Principles & Rules

Rule #	Mandatory Architectural Principle Statement
1	React is presentation / UI only — no direct database calls or business logic.
2	Next.js BFF is the browser-facing session boundary — manages HttpOnly cookies and session storage.
3	YARP is the common security gateway through which all protected PWB traffic passes.
4	MPOEmployeeHub owns authentication, SSO, and identity token lifecycle end-to-end.
5	Layer 5 owns all PWB business functionality — APIs, business rules, and integrations.
6	Layer 6 owns persistent data storage — business data, transactions, and audit records.
7	All service-to-service communication must enforce HTTPS encryption.
8	Role-based authorization must be validated before protected business operations execute.
9	Backend complexity must be hidden from the browser client via the BFF layer.
10	Exact low-level class designs, DB schemas, and API JSON payloads belong to LLD.